Algorithm 1 Quickhull

Require: Set P of points $p_n(x, y)$

Ensure: Ordered list H of vertices from P forming the convex hull

function FIND_HULL(A, B, S)
    if $S = \emptyset$ **then**
        return $[\,]$
    end if
    $C \leftarrow \arg\max_{p \in S} \overrightarrow{AB} \times \overrightarrow{Ap}$
    $S_1 \leftarrow \{p \in S \mid \overrightarrow{AC} \times \overrightarrow{Ap} > 0\}$
    $S_2 \leftarrow \{p \in S \mid \overrightarrow{CB} \times \overrightarrow{Cp} > 0\}$
    return FIND_HULL$(A, C, S_1) +\!\!+ [C] +\!\!+$ FIND_HULL(C, B, S_2)
end function

if $P = \emptyset$ **then**
    return $[\,]$
end if
$A \leftarrow \arg\min_{p \in P}(p_x)$
$B \leftarrow \arg\max_{p \in P}(p_x)$
if $A = B$ **then**
    return $[A]$
end if
$S_1 \leftarrow \{p \in P \mid \overrightarrow{AB} \times \overrightarrow{Ap} > 0\}$
$S_2 \leftarrow \{p \in P \mid \overrightarrow{AB} \times \overrightarrow{Ap} < 0\}$
$H \leftarrow [A] +\!\!+$ FIND_HULL$(A, B, S_1) +\!\!+ [B] +\!\!+$ FIND_HULL(B, A, S_2)
return H